expect search boxes to provide it by default.

The usual autocomplete workflow is the following: a user on the client side (typically a browser) starts typing in a few letters, and the autocomplete search box shows a few options that starts with the characters already typed (ring a bell? Sounds like “all keys with prefix”?). If the set of possible values that could be inserted in the search box was static and small, then it could be transmitted to the client together with the page, cached, and used directly on the client.

This, however, is usually not the case, since the sets of possible values are normally large, and they might change in time, or even be dynamically queried.

So, in real-world applications the client usually sends (asynchronously) a REST request to the server with the characters typed so far.

The application server keeps a trie (or more likely a Patricia tree) with the valid entries, searches for the strings starting with what was inserted so far and, that’s a valid prefix, returns a certain number of entries in this string's subtree.

When the response comes back to the client, it simply shows the list of results from the server as suggestions.

It's not strictly required neither that the request is performed through a REST endpoint nor that it is sent asynchronously. Yet, the former indicates a clean design, and without the latter the autocomplete feature would make little sense.